

PH.D. APPLIED QUALIFYING EXAM 2023 (TAKE HOME)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. The exam consists of two problems, both of which you need to answer. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have questions, please email Prof. Han Xiao (hxiao@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

- (1) Create a folder containing the following:
 - (a) A word, html or pdf file containing your answers to the two problems, including all tables, figures and Appendices (this may be generated from an R markdown file)
 - (b) An R (or python) file containing the codes used to generate your results, or an R markdown file from which your output html or pdf is generated.
 - (c) A word or a text file that gives a summary of the codes submitted (not required if you submit an R markdown file).
- (2) Name the folder **yourlastname_yourfirstname_Appliedexam2023**. Zip this folder and email it to hxiao@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 7, Friday.

Problem 1

1. OVERVIEW

In this problem, you will be acting as statistical consultants for a company that manufactures consumer products, and answer specific questions asked by the company management, who do not care much about the novelty of statistical methodology you use, as long as their questions are answered. However, your methodology (and its statistical validity) is accountable to your despicable bosses – Ying Hung and Tirthankar Dasgupta. There is no need to restrict your answers to what was covered in Stat 596 and Stat 597. Use your instincts, your previous training in data analysis, and focus on answering the questions of interest using correct statistical methodology.

2. BACKGROUND

The company has conducted a product marketing experiment before launching a new dish-washing detergent. Two “test” detergents were manufactured. A total of 549 households were chosen as the experimental units for this study, and the two test products were assigned randomly to these households. Each household received by mail one of two test products, used the product and returned a survey regarding the product’s performance. The survey data is available in the file `market_survey_data_mod.csv`.

The first column (A) of the file contains the household identification number ranging from 1 to 549, and the second column (B) contains the test product number (nominal variable with values 1–2) assigned to the household. The Columns C through K marked in green contain the outcomes or the response variables and are answers to the following questions on a scale of 1–3, with 1 being the worst and 3 being the best.

- (1) Column C: Does the product provide value for money? (1:Poor, 3:Excellent)
- (2) Column D: How would you rate the product overall? (1:Poor, 3:Excellent)
- (3) Column E: How do you rate the product with respect to its performance (cleaning dishes)? (1:Poor, 3:Excellent)
- (4) Column F: Do you like the scent (perfume) of the product? (1:Poor, 3:Excellent)
- (5) Column G: Does use of the product irritate your hand? (1:A lot, 3:Not at all)
- (6) Column H: Did you like the color of the product? (1: Not at all, 2: Very much)
- (7) Column I: Are you satisfied with the packaging? (1:Not at all, 3: Extremely satisfied)
- (8) Column J: How much effort did you spend in dispensing the detergent? (1:A lot of effort, 3: It was very easy)
- (9) Column K: Did you think the package was filled as per your expectation: (1:Poor, 3:Excellent)

Note that the responses in columns C and D represent the users' overall evaluation of the product, whereas the other columns represent their views on specific aspects of the product. The outcome in column D (overall rating) is the one of primary interest to the management, although they are also interested in the other outcomes.

The survey also collected covariate information that are reported in the blue columns L through R. Descriptions of these columns are given below:

- (1) Column L: Do you place the product directly under a running stream of water? (Yes/No)
- (2) Column M: How do you wash dishes (1= only by hand; 2=most of them by hand but some in machine; 3=some by hand but most of them in machine; 4= equally by hand and machine; 5=only in machine)
- (3) Column N: Have you been using brand X of dishwashing detergent during the past six months (prior to this survey)? (Yes/No)
- (4) Column O: Household income (nominal variable with 11 levels)
- (5) Column P: Age of the person primarily responsible for washing dishes in the house (nominal variable with 6 levels, each level measuring an interval, e.g. 15-24.)
- (6) Column Q: Number of hand washes per week (discrete variable)
- (7) Column R: Hispanic? (Yes/No)

3. QUESTIONS

- (1) There are many missing observations in the Excel file, both with respect to outcomes and covariates. Can you quickly compile these missing values by questions and create a visualization to see whether there are specific questions that have lower response rates?
- (2) Is it possible to predict the overall performance measures (columns C and D) from the remaining performance measures (columns E to K)? That is, do the ratings on specific features like performance, perfume, color, packaging, effect on hand, ease of dispensing, extent to which the package is filled, affect the overall performance measures? If so, fit appropriate models (individual model for each overall performance measure) for each product, and compare them.
- (3) How well can you predict the overall performance measures for each product from the covariates? Fit appropriate statistical models (individual model for each overall performance measure) and compare them for the two products.
- (4) Does it appear that the two test products differ significantly with respect to the response of primary interest (overall rating)? If they do, which is better?

Write a report to the management answering the above four questions. Feel free to use any statistical methodology you want to, but state your analysis methods clearly and unambiguously. You can use any R package you want to and/or write your own codes. Submit your R codes so that your

results can be replicated. You can also create a single R markdown file with your codes and the html or pdf file generated from it.

Problem 2

Assume that a deep learning algorithm involving two tuning parameters, x_1 and x_2 , is given to perform image recognition. It is also assumed that the underlying performance of these two tuning parameters in terms of prediction error can be described by the following deterministic function:

$$y = [1 + (x_1 + x_2 + 1)^2(19 - 14x_1 + 3x_1^2 - 14x_2 + 6x_1x_2 + 3x_2^2)] \times \\ [30 + (2x_1 - 3x_2)^2(18 - 32x_1 + 12x_1^2 + 48x_2 - 36x_1x_2 + 27x_2^2)],$$

where y is the average prediction error, x_1 and x_2 are evaluated on a square defined by $x_i \in [-2, 2]$ for $i = 1, 2$.

- (1) In reality, this relationship is unknown. Instead we can train the deep learning algorithm on 50 different tuning parameters settings sampled from the square, denoted by (x_{1j}, x_{2j}) and collect the corresponding y , denoted by y_j , from the given relationship, where $j = 1, \dots, 50$. Based on this idea, conduct a simulation to generate training data on the tuning parameters and the prediction error with sample size 50. Visualize the training data by plotting
 - (1a) the true deterministic function v.s. x_1 with $x_2 = 2$,
 - (1b) the true deterministic function v.s. x_2 with $x_1 = -2$,
 - (1c) the true deterministic function (3D plot) with the training data labelled.
- (2) Fit a surrogate model based on the training data by assuming the data are realizations from a Gaussian process $Y = \mu + Z(x)$, where μ is unknown and $Z(x)$ is a Gaussian process with mean 0 and covariance function defined by $\text{cov}(Y((x+h), Y(x)) = \sigma^2 \phi(h; \theta)$. Define the correlation function $\phi(h; \theta)$ by

$$\phi(h; \theta) = \exp(-\theta h^2).$$

Write down the theoretical expressions of MLEs for parameters μ , θ , and σ^2 . Based on the 50 observations, what are your estimated MLEs?

- (3) Based on (2), write down the best linear unbiased predictor (BLUP) and the (plugin) predictive distribution of y at $x_1 = x_1^*$ and $x_2 = x_2^*$?
- (4) Based on the predictive distribution described in (3), plot the BLUP and the 95% predictive confidence bands on (1a) and (1b).
- (5) Based on the BLUP found in (3), what is the best tuning parameter setting that minimizes y ?
- (6) If you are allowed to add five more training data points, what are the parameter settings you would like to try? Provide a brief justification behind selection of these points.
- (7) Based on the same set of data, fit a spline-based model and find the corresponding best tuning parameter settings that minimize y .